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## Camera calibration method based on vehicle localization


### Abstract

The present disclosure provides a camera calibration method based on vehicle localization, mainly considering that a vehicle is not of an ideal rigid structure, and shaking of a vehicle body during operation, uneven road surfaces, or different loads will cause variations in extrinsic parameters of a camera system, resulting in perceptual information errors. Therefore, through perceptual information from a fused image, an inertial measurement unit, a speedometer, and the like, combined with the road and marking information provided by a high-definition map, the relative motion between the vehicle body and a vehicle chassis is simultaneously described with a mass-spring-damper model, so as to complete the determination of six-degree-of-freedom positions of the vehicle body and the vehicle chassis, and finally the extrinsic parameters of cameras are immediately calibrated with the vehicle chassis as reference coordinates.



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## References Cited


**U.S. PATENT DOCUMENTS**

| Patent No. | Issued Date | Patentee Name | U.S. Cl. | CPC |
|---|---|---|---|---|
| 6594600 | 12/2002 | Arnoul | 702/94 | G06T 7/85 |
| 8150143 | 12/2011 | Chen | 345/178 | G06T 7/80 |
| 8842181 | 12/2013 | Kosaki | N/A | N/A |
| 10565736 | 12/2019 | Oba et al. | N/A | N/A |
| 10694175 | 12/2019 | Bovyrin et al. | N/A | N/A |
| 10798303 | 12/2019 | Camyre | N/A | G01S 17/06 |
| 10919458 | 12/2020 | Gupta et al. | N/A | N/A |
| 11428541 | 12/2021 | Khalili et al. | N/A | N/A |
| 2010/0165116 | 12/2009 | Hsieh | 348/187 | H04N 5/2224 |
| 2010/0238291 | 12/2009 | Pavlov | 348/148 | G06T 7/73 |
| 2010/0315505 | 12/2009 | Michalke | 348/118 | G06T 7/251 |
| 2013/0293717 | 12/2012 | Zhang et al. | N/A | N/A |
| 2017/0134722 | 12/2016 | Okouneva | N/A | N/A |
| 2020/0329215 | 12/2019 | Tsunashima | N/A | N/A |
| 2021/0327092 | 12/2020 | Jiang et al. | N/A | N/A |
| 2022/0381557 | 12/2021 | Rogers et al. | N/A | N/A |

**FOREIGN PATENT DOCUMENTS**

| Patent No. | Application Date | Country | CPC |
|---|---|---|---|
| 104019829 | 12/2013 | CN | N/A |
| 110967038 | 12/2019 | CN | N/A |
| 113658252 | 12/2020 | CN | N/A |
| 115943636 | 12/2022 | CN | N/A |
| I424259 | 12/2013 | TW | N/A |
| 2023232318 | 12/2022 | WO | N/A |

**OTHER PUBLICATIONS**

Longitudinal Improvement for Self-Localization Based on Mono-Camera and Traffic Signs, 2018. cited by applicant

Semantic Segmentation-Based Lane Level Localization Using Around View Monitoring System, 2019. cited by applicant

A Light-Weight Semantic Map for Visual Localization towards Autonomous Driving, 2021. cited by applicant

Monocular Vehicle Self-localization method based on Compact Semantic Map, 2018. cited by applicant

LaneMatch: A Practical Real-Time Localization Method Via Lane-Matching, 2022. cited by applicant

Lane-level localization system using surround-view cameras adaptive to different driving conditions, 2020. cited by applicant
Advances in Real-Time Online Vehicle Camera Calibration via Road Line Markings Parallelism Enforcement, 2022. cited by applicant

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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION
(1) The present application is based on, and claims priority from, Taiwan (International) Application Serial Number 112148663, filed on Dec. 14, 2023, the disclosure of which is hereby incorporated by reference herein in its entirety.
TECHNICAL FIELD
(2) The present disclosure belongs to a camera calibration method based on vehicle localization, and relates to a dynamic camera calibration method applying high-definition map information and localization, and is used for achieving the function of combining visual-based localization and dynamic camera calibration.
BACKGROUND
(3) An operation mode of a self-driving vehicle (an autonomous vehicle) is mainly to configure various sensors in different positions of the self-driving vehicle according to different application requirements, enabling the sensors to detect various driving information when the self-driving vehicle travels, to provide reference for a self-driving system to plan control commands, and then to manipulate the self-driving vehicle to stably travel.
(4) However, the self-driving vehicle is actually of a non-rigid structure composed of multiple parts, the significance of which lies in the fact that when the self-driving vehicle moves, all the parts of the self-driving vehicle will relatively move, and the relative displacement of the different parts may cause the different sensors thereon to shake. For example, depending on vehicle models, a head portion of a trailer will sway significantly during traveling; and a cargo van will have different body heights due to different loads. In addition, uneven road surfaces may also cause a vehicle body to sway, which may result in sensing errors of the sensors.
(5) Such shaking and uneven road surfaces cause the sensors to offset from 6D position parameters, a.k.a. extrinsic parameters, set by the self-driving system, resulting in errors in detecting motion states of surrounding objects by the self-driving vehicle, such as relative distances and speeds of the surrounding objects to the self-driving vehicle. The errors in the estimation of positions of the surrounding objects of the self-driving vehicle are prone to leading to the inability of the self-driving system to calculate the most appropriate control commands, which may result in a relatively large speed and acceleration variation of the self-driving vehicle due to large variations in the states of the objects, or even collisions between the self-driving vehicle and other objects.
(6) Taking an existing self-driving system or ADAS as an example, it usually uses cameras assembled on the vehicle body to obtain the positions of the surrounding objects, which serve as extrinsic parameters set by the self-driving system through an image-based object detection and localization method. However, because the body of the vehicle is actually connected to a vehicle chassis through a suspension system rather than a rigid structure, extrinsic parameters of six-degree-of-freedom positions of the cameras configured on the vehicle body relative to a camera at a reference point of coordinates of the vehicle chassis are susceptible to variations due to the swaying of the vehicle body, the different loads, and the tilting of the road surfaces, which may result in

variations of rotation angles and displacements of the cameras relative to the vehicle chassis, leading to errors in distances between the objects detected based on images and the vehicle and, consequently, possibly causing the malfunctioning of the self-driving system or the ADAS.

SUMMARY

(7) An embodiment of the present disclosure provides a camera calibration method based on vehicle localization, which combines vehicle chassis localization with multi-sensor fusion and visual-based vehicle body localization, coupled with operation of dynamic model iteration of a vehicle body and a vehicle chassis, to complete determination of six-degree-of-freedom positions of the vehicle body and vehicle chassis, and to calibrate extrinsic parameters of cameras with the vehicle chassis as reference coordinates.

(8) An embodiment of the present disclosure provides the vehicle chassis localization with multi-sensor fusion referred to obtaining speed information of the chassis of a vehicle, a speed and acceleration of a three-dimensional displacement of the vehicle chassis, and an angular speed and angular acceleration of a three-dimensional rotation angle of the vehicle chassis, to estimate a current initial six-degree-of-freedom position of the chassis of the vehicle.

(9) An embodiment of the present disclosure provides the image-based vehicle body localization referred to using an surround view cameras to obtain an around view image of an vicinity of the body of the vehicle, detecting an image object feature point set in the image and vanishing points of all the cameras of the surround view cameras, and matching the same with a semantic map feature point set of the vicinity of the vehicle, to calculate six-degree-of-freedom positions of all the cameras.

(10) An embodiment of the present disclosure provides iterative approach of a relative motion model of the vehicle body and the vehicle chassis referred to estimating a three-dimensional relative displacement between the vehicle body and the vehicle chassis by using a mass-spring-damper model, iteratively and finely adjusting the six-degree-of-freedom position of the vehicle chassis with reference to a road surface normal vector and the rotation angle of the vehicle chassis, to complete the determination of the six-degree-of-freedom positions of the vehicle body and the vehicle chassis, and finally calibrating the extrinsic parameters of the cameras with the vehicle chassis as reference coordinates.

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## Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a workflow chart of a dynamic camera calibration method according to an embodiment of the present disclosure;

(2) FIG. **2** is a front view of a structure of a vehicle sensor configuration according to an embodiment of the present disclosure, with a vehicle model shown for reference only; and

(3) FIG. **3** is a top view of a structure of a vehicle sensor configuration according to an embodiment of the present disclosure, with a vehicle model shown for reference only.

DESCRIPTION OF SIGNS OF MAIN COMPONENTS

(4) **100** vehicle **110** vehicle body **120** vehicle chassis **210** speedometer **220** inertial measurement unit **230** mass-spring-damper model **300** surround view cameras **310** camera **400** high-definition map information

DETAILED DESCRIPTION

(5) Below, exemplary embodiments of the present disclosure will be described in detail with reference to accompanying drawings so that the features of the present disclosure can be more readily understood by those skilled in the art.

(6) Reference is made to FIG. **1**, which is a camera calibration method based on vehicle localization provided by the present disclosure. A system architecture of the dynamic camera

calibration method, as shown in FIGS. **2** and **3**, includes a vehicle **100**. The vehicle **100** has an upper half portion of a vehicle body **110** and a lower half portion of a vehicle chassis **120**. The vehicle chassis **120** is provided with a speedometer **210** and an inertial measurement unit **220**, and the vehicle body **110** is provided with an surround view cameras **300**. The surround view cameras **300** includes a plurality of cameras **310**, which are dispersed on the periphery of the vehicle body to obtain surrounding view images. In addition, a mass-spring-damper model **230** is connected between the vehicle body **110** and the vehicle chassis **120**, and the mass-spring-damper model **230** calculates the variation in elongation of the mass-spring-damper model with the vehicle chassis **120** as reference coordinates, so as to estimate a three-dimensional relative displacement between the vehicle body **110** and the vehicle chassis **120**.
(7) With the above structural composition, the dynamic camera **310** correction method at least includes the following steps: Step A: through localization of the vehicle chassis **120** with multi-sensor (e.g. the speedometer **210**, and the inertial measurement unit **220**) fusion, estimate an initial six-degree-of-freedom position, speed and acceleration of the vehicle chassis **120**; Step B: through visual-based localization of the vehicle body **110**, apply a semantic map feature points to match with detected features of the surround view cameras **300**, so as to calculate determination of six-degree-of-freedom positions of all the cameras **310**; and Step C: through iterative approach of a relative motion model of the vehicle body **110** and the vehicle chassis **120**, complete determination of six-degree-of-freedom positions of the vehicle body **110** and vehicle chassis **120**, and finally calibrate extrinsic parameters of all the cameras **310** with the vehicle chassis **120** as reference coordinates.
(8) Based on the above, the detailed workflow of the present disclosure is described as follows.
(9) As for the localization of the vehicle chassis **120** with multi-sensor (e.g. the speedometer **210**, and the inertial measurement unit **220**) fusion (step A), it is required to use the speedometer **210** to sense speed information of the vehicle chassis **120** and use the inertial measurement unit **220** to obtain a speed and acceleration of a three-dimensional displacement of the vehicle chassis **120**, and an angular speed and angular acceleration of a three-dimensional rotation angle of the vehicle chassis **120**, to estimate the current initial six-degree-of-freedom position of the chassis **120** of the vehicle **100** (step A**01**).
(10) As for the visual-based localization of the vehicle body **110** (step B), the surround view cameras **300** are used to obtain surround view images of the body **110** of the vehicle **100**. The image-detected stop lines, pedestrian crossings, other road markers, and image features (step B**01**) composes an image object feature point set (step B**02**). Vanishing points of all the cameras **310** of the surround view cameras **300** are calculated from the feature point set (step B**03**). The Initial estimation values of rotation angles of all the cameras **310** are calculated based on the vanishing points (step B**04**) and the feature point set match with the semantic map feature point set of the vicinity of the vehicle **100** queried from high-definition map information **400** to calculate the six-degree-of-freedom positions of all the cameras **310** (step B**05**).
(11) Then, the iterative approach of the relative motion model of the vehicle body **110** and the vehicle chassis **120** is conducted (step C), the three-dimensional relative displacement between the vehicle body **110** and the vehicle chassis **120** is estimated by using the mass-spring-damper model **230**, the six-degree-of-freedom position of the vehicle chassis **120** is iteratively and finely adjusted with reference to the road surface normal vector and the rotation angle of the vehicle chassis **120** (step C**01**), to complete the determination of six-degree-of-freedom positions of the vehicle body **110** and the surround view cameras **300** thereon (step C**02**) as well as the determination of the six-degree-of-freedom position of the vehicle chassis **120** (step C**03**), and finally, the extrinsic parameters of all the cameras **310** of the surround view cameras **300** are calibrated with the vehicle chassis **120** as reference coordinates (step C**04**).
(12) In the foregoing method, the current initial six-degree-of-freedom position of the vehicle chassis **120** may further be estimated by using speed and acceleration information of the vehicle

**100** calculated by the speedometer **210** and the inertial measurement unit **220**, and taking a position of the vehicle chassis **120** at a previous sampling time point as a reference point (step A**03**) to calculate a position of the vehicle chassis **120** at a current sampling time point. Similarly, as for estimation of the six-degree-of-freedom positions of all the cameras **310**, the six-degree-of-freedom positions of all the cameras **310** at a current sampling time point may also be calculated by taking a position of the vehicle body **110** at a previous sampling time point as a reference point (step B**06**) when the image object feature point set and the semantic map feature point set are matched.
(13) Moreover, the road surface normal vector is generated by querying the high-definition map (HD Map) information **400** according to the position of the vehicle chassis **120**.
(14) In addition, the semantic map feature point set of the vicinity of the vehicle **100** is generated by querying the high-definition map information **400** with the position of the vehicle chassis **120**, and includes a set of a road marking edge and a traffic sign edge.
(15) Besides, the vanishing points of all the cameras **310** of the surround view cameras **300** and the image object feature point set are calculated according to a plurality of image features in the around view image.
(16) Furthermore, as for the six-degree-of-freedom extrinsic parameters of all the cameras **310** of the surround view cameras **300**, pitch and yaw angles of each of the cameras **310** are calculated according to the vanishing points of the all cameras **310** in the surround view cameras **300**; the all cameras **310** are configured on the vehicle body **110** and are in rigid connection with the vehicle body **110**, and the pitch and yaw angles of the all cameras **310** are used for determining an initial value of a three-dimensional rotation angle of the vehicle body **110**; and matching is conducted according to the semantic map feature point set of the vicinity of the vehicle **100** and the image object feature point set detected by the cameras **310** to calculate the six-degree-of-freedom positions of the cameras **310** and the six-degree-of-freedom position of the vehicle body **110**.
(17) In addition, the six-degree-of-freedom position of the vehicle body **110** and the six-degree-of-freedom position of the vehicle chassis **120** are iterated by, according to an initial value of the six-degree-of-freedom position of the vehicle body **110**, an initial value of the six-degree-of-freedom position of the vehicle chassis **120**, an initial value of the speed and acceleration of the vehicle chassis **120**, and the road surface normal vector obtained by querying the high-definition map according to a position of the vehicle chassis **120**, using the mass-spring-damper model to describe relative motion of the vehicle body **110** and the vehicle chassis **120**, dynamic values of the mass-spring-damper model **230**, including a position, a speed and an acceleration are continuously corrected, until an external force applied to the vehicle body **110** and the vehicle chassis **120** is approximate to a dynamic variation of the mass-spring-damper model and the six-degree-of-freedom position of the vehicle body **110** and the six-degree-of-freedom position of the vehicle chassis **120** are generated, and the six-degree-of-freedom extrinsic parameters of the cameras **310** of the surround view cameras **300** are calculated.
(18) The embodiment disclosed above at least has the following characteristics.
(19) As for a self-driving system or an ADAS, the mass-spring-damper model is introduced to describe the relative displacement between the vehicle body and the vehicle chassis, and the extrinsic parameters of the cameras are immediately calibrated with the vehicle chassis as reference coordinates, so as to solve the problem that due to shaking of the vehicle body during operation, uneven road surfaces, or different loads, the rotation angles and displacements of the cameras relative to the vehicle chassis are varied, thereby resulting in perception and localization errors, and to achieve the function of combining visual-based localization and dynamic camera calibration.
(20) While the embodiments of the invention have been set forth for the purpose of disclosure, modifications of the disclosed embodiments of the invention as well as other embodiments thereof may occur to those skilled in the art. Accordingly, the appended claims are intended to cover all embodiments which do not depart from the spirit and scope of the invention.

## Claims

1. A camera calibration method based on vehicle localization, comprising: obtaining a speed and acceleration of a three-dimensional displacement of a vehicle chassis and an angular speed and angular acceleration of a three-dimensional rotation angle of the vehicle chassis to estimate a current initial six-degree-of-freedom position of the chassis of a vehicle; disposing an surround view cameras comprising a plurality of cameras on the vehicle, using the surround view cameras to obtain an around view image of an vicinity of a body of the vehicle, detecting an image object feature point set in the around view image and vanishing points of all the cameras of the surround view cameras, and matching with a semantic map feature point set of the vicinity of the vehicle, to calculate six-degree-of-freedom positions of all the cameras; connecting a mass-spring-damper model between the vehicle body and the vehicle chassis, and estimating, by the mass-spring-damper model, a three-dimensional relative displacement between the vehicle body and the vehicle chassis with the surround view cameras as a reference point; iterating the six-degree-of-freedom position of the vehicle chassis according to a road surface normal vector and the rotation angle of the vehicle chassis, to complete determination of six-degree-of-freedom positions of the vehicle body and the vehicle chassis; and calibrating the six-degree-of-freedom positions of the cameras with the vehicle chassis as reference coordinates.
2. The camera calibration method based on vehicle localization according to claim 1, wherein the current initial six-degree-of-freedom position of the vehicle chassis is estimated by using speed and acceleration information of the vehicle calculated by a speedometer and an inertial measurement unit, and taking a position of the vehicle chassis at a previous sampling time point as a reference point to calculate a position of the vehicle chassis at a current sampling time point.
3. The camera calibration method based on vehicle localization according to claim 1, wherein the road surface normal vector is generated by querying high-definition map information according to a position of the vehicle chassis.
4. The camera calibration method based on vehicle localization according to claim 1, wherein the semantic map feature point set of the vicinity of the vehicle is generated by querying high-definition map information with a position of the vehicle chassis, and comprises a set of a road marking edge and a traffic sign edge.
5. The camera calibration method based on vehicle localization according to claim 1, wherein the vanishing points of all the cameras of the surround view cameras and the image object feature point set are calculated according to a plurality of image features in the surround-view images.
6. The camera calibration method based on vehicle localization according to claim 1, wherein as for the six-degree-of-freedom positions of all the cameras of the surround view cameras, pitch and yaw angles of each of the cameras are calculated according to the vanishing points of the plurality of cameras in the surround view cameras; the plurality of cameras are configured on the vehicle body and are in rigid connection with the vehicle body, and the pitch and yaw angles of the plurality of cameras are used for determining an initial value of a three-dimensional rotation angle of the vehicle body; and matching is conducted according to the semantic map feature point set of the vicinity of the vehicle and the image object feature point set detected by the cameras to calculate the six-degree-of-freedom positions of the cameras and the six-degree-of-freedom position of the vehicle body.
7. The camera calibration method based on vehicle localization according to claim 1, wherein the six-degree-of-freedom position of the vehicle body and the six-degree-of-freedom position of the vehicle chassis are iterated by, according to an initial value of the six-degree-of-freedom position of the vehicle body, an initial value of the six-degree-of-freedom position of the vehicle chassis, an initial value of the speed and acceleration of the vehicle chassis, and the road surface normal vector obtained by querying a high-definition map according to a position of the vehicle chassis, using a

mass-spring-damper model to describe relative motion of the vehicle body and the vehicle chassis, dynamic values of the mass-spring-damper model, comprising a position, a speed and an acceleration are continuously corrected, until an external force applied to the vehicle body and the vehicle chassis is approximate to a dynamic variation of the mass-spring-damper model and the six-degree-of-freedom position of the vehicle body and the six-degree-of-freedom position of the vehicle chassis are generated, and the six-degree-of-freedom positions of the cameras of the surround view cameras are calculated.